

Embedded Panels Maximise Solar Energy Output

High-voltage embedded solar panels wired in parallel lower the installed cost of the solar power system and improve the energy harvesting efficiency by up to 30 per cent, while eliminating the design and installation constraints inherent in conventional series/parallel string architecture

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Solar photovoltaic arrays have tremendous potential to alter the economics and environmental impact of electrical power generation. However, despite large innovation in the technology of photovoltaic modules, the system design had so far remained largely unchanged for over a decade.

Even today most solar arrays are composed of 'strings' of matched panels, wired in series. This conventional approach is fundamentally vulnerable—when individual panels experience minor variations in the output due to shadow, dust or partial failure, these disrupt the output of the entire string and the entire array, sharply cutting overall energy harvest of the array.

Parallel solar architecture addresses this inherent vulnerability when the array is built out of high-voltage embedded solar panels (ESPs). The digital signal processor (DSP) controlled solar energy processing inside a photovoltaic (PV) panel delivers regulated high-voltage DC at any voltage up to 300V DC. The embedded DSP controller has

a special algorithm that does maximum peak power tracking (MPPT) from the panel to ensure maximum possible extraction of solar energy and its delivery at a constant high voltage. Each ESP is designed for N+1 parallel operation with similar ESPs. This allows the array to have a single high-voltage bus bar in its parallel architecture.

Each ESP is thus a fully independent, highly efficient contributor of power to the array—just like a power plant on a grid. This gives designers the flexibility in PV panel location, orientation and overall system size. Installers have a safe, simple wiring scheme with easy expansion and per-panel monitoring. And most importantly, system operators improve their expected output and payback.

What are ESPs?

An ESP is a conventional photovoltaic solar panel but with an embedded DSP-controlled energy processing module designed and tested to work at high operating temperatures (up to 80°C) and packaged for environmental endurance as long as the life of the

panel. Besides tracking the peak-power operating point for the solar panel, ESPs output stable high-voltage DC at 95 per cent peak efficiency. These are available with both poly-crystalline and mono-crystalline solar cells. Their embedded DSP helps to continuously monitor the energy delivered by each ESP.

Constant voltage/constant power

ESPs deliver constant peak power irrespective of the battery condition if storage batteries are connected across their outputs. These pump in constant peak power into batteries from fully discharged state till they are fully charged. This ideally suits solar systems that store energy during sunny hours and deliver power at night. Next day, the discharged battery gets quickly charged at constant power.

Why parallel solar?

PV solar system designers have always faced four technological problems that cause drop in the energy yield besides adding to the cost due to system complexities in panel utilisation:

1. The low voltage output of individual panels made it necessary to connect several PV panels in series to derive high voltage from an array to feed high-power inverters. In addition, stringing panels in series required them to be selected and matched in order to get the maximum energy.

2. Problems in many sites were caused by surrounding trees casting shadows during parts of a day. Shad-

